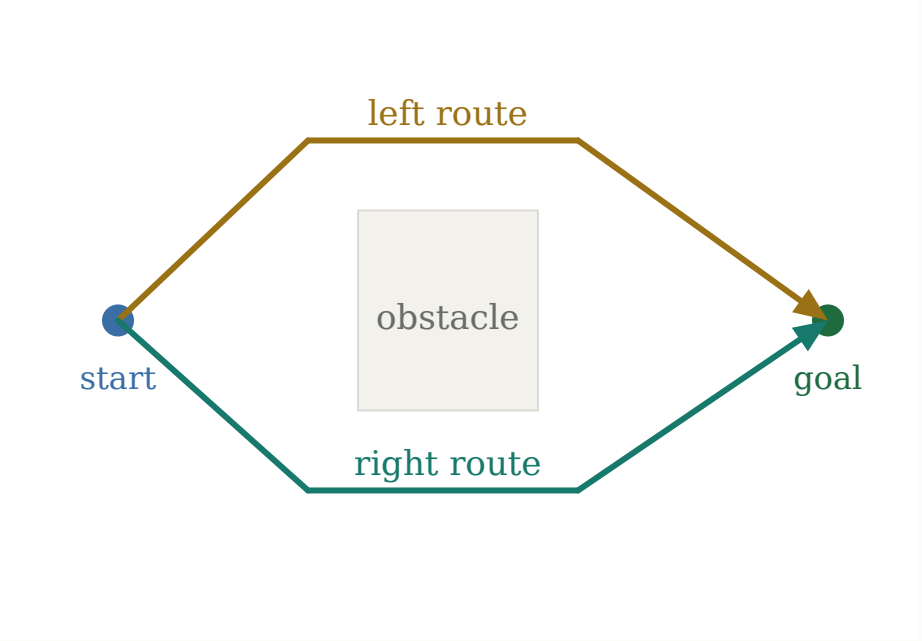
MOTIVATION

Information Theory, Sampling and Density

How should a robot compare uncertain predictions?

Two safe routes can have different probabilities. A useful model must represent both, and we need evidence that it does.



Lecture 9 · November 11, 2026 · Mathematics for Robotics · ROB-GY 6013
NYU Tandon · Jing Zhang · Fall 2026

Goodfellow et al., Deep Learning (2016), §3.13

ROADMAP

The reasoning path

One robot problem, several different questions

Score the routes and ask what the sensor tells us. Use trials to estimate unknown quantities, then model continuous action groups. Stable arithmetic supports the probability calculations.

Routes and probabilities
Score the model

Hidden state and reports
Evaluate the sensor

Observed trials
Estimate unknown quantities

Continuous actions
Find action groups

Model scores
Compute probabilities safely

Goodfellow et al., Deep Learning (2016), §3.13

RECALL

Probability recall

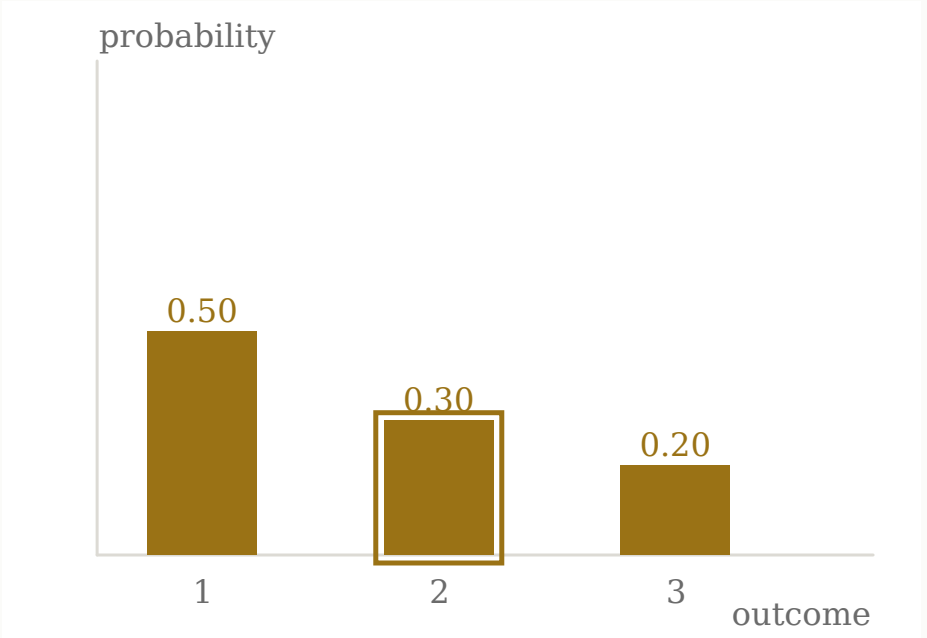
The distribution and its samples are different objects

For finite outcomes, expectation is a probability-weighted average. The source probabilities satisfy $p(x) \geq 0$ and $\Sigma_x p(x) = 1$. The graph shows three outcomes.

$$\mathbb{E}_p[g(X)] = \sum_x p(x) g(x)$$

`X,x` random outcome and possible value

`p,g` probability mass function and measured quantity



Goodfellow et al., Deep Learning (2016), §3.8; original example

DEFINITION

Surprisal

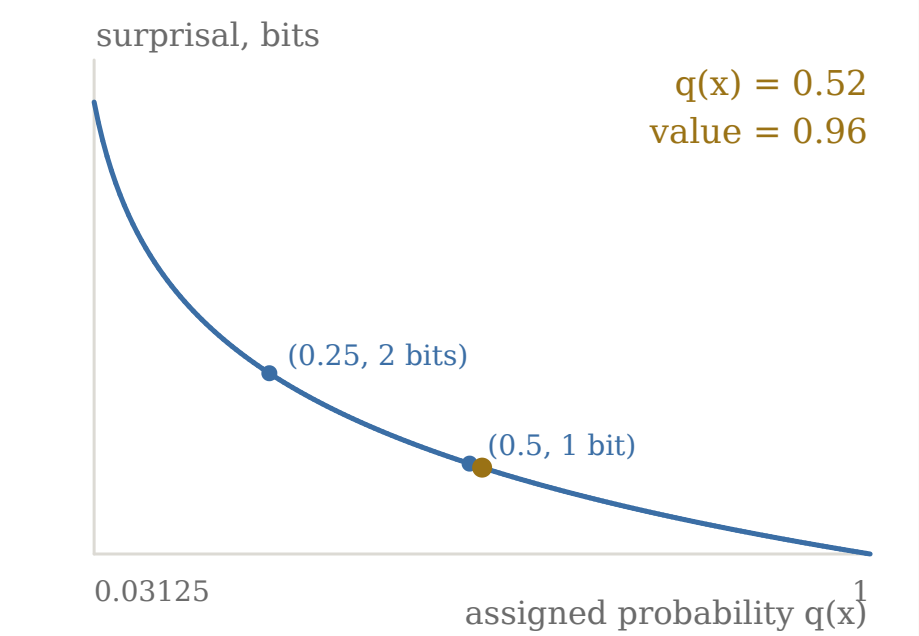
A confident mistake should cost more

Surprisal is the negative logarithm of the probability assigned to an observed event. Here $0<q(x)\leq 1$ and the log base is 2.

$$I_q(x) = -\log_2 q(x)$$

`q(x)` predicted probability of the observed event

`I_q(x)` surprisal, measured in bits



Goodfellow et al., Deep Learning (2016), §3.13

DERIVATION

Surprisal

Independent evidence adds in log space

For independent events, their joint probability is the product.

$$-\log_2(q_1 q_2) = -\log_2 q_1 - \log_2 q_2$$

`q_1,q_2` probabilities of independent events

Goodfellow et al., Deep Learning (2016), §3.13

Surprisal

Independent evidence adds in log space

Factor the independent joint probability

Independent outcomes x_1 and x_2 have positive masses $q_1 = q(x_1)$ and $q_2 = q(x_2)$.

1. $q(x_1, x_2) = q_1 q_2$ by independence.
2. $I_q(x_1, x_2) = -\log_2(q_1 q_2)$ by the definition of surprise.
3. $-\log_2(q_1 q_2) = -\log_2 q_1 - \log_2 q_2$ by the log-product rule.

Goodfellow et al., Deep Learning (2016), §3.13

DEFINITION

Entropy

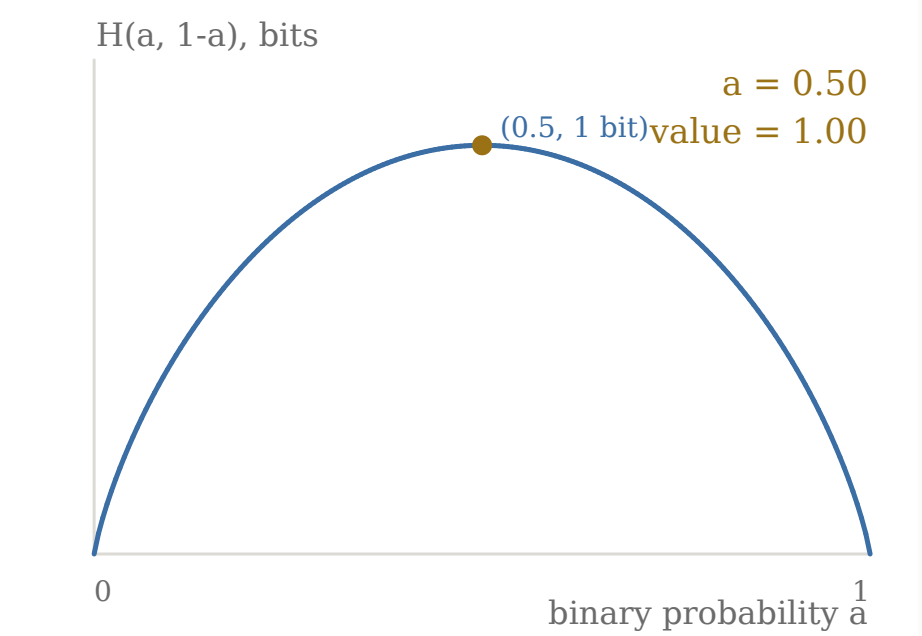
Average surprise under the true distribution

Entropy is the expected surprisal of an outcome under its own distribution. The curve shows binary distributions (a,1−a).

$$H(p) = -\sum_{x} p(x) \log_2 p(x)$$

`p(x)` probability of outcome x

`H(p)` entropy in bits; use 0 log 0 = 0



Goodfellow et al., Deep Learning (2016), §3.13

EXAMPLE

Entropy

A concrete prefix code makes the units visible

COURSE-CREATED WORKED EXAMPLE

A prefix code has no complete codeword that is the beginning of a different codeword. Pair (1/2, 1/4, 1/8, 1/8) with (0, 10, 110, 111).

$$L = \tfrac{1}{2}(1) + \tfrac{1}{4}(2) + \tfrac{1}{8}(3) + \tfrac{1}{8}(3)$$

L expected codeword length in bits

symbol	probability	codeword
A	1/2	0
B	1/4	10
C	1/8	110
D	1/8	111

Shannon (1948), §§6,9–12; original course examples

EXAMPLE

Entropy

The code length equals entropy in this example

COURSE-CREATED WORKED EXAMPLE

For p=(1/2, 1/4, 1/8, 1/8), the ideal lengths $-\log_2 p(x)$ are 1, 2, 3, 3 bits.

$$H(p) = \tfrac{1}{2}\log_2 2 + \tfrac{1}{4}\log_2 4 + \tfrac{1}{4}\log_2 8 = 1.75$$

`p` (1/2, 1/4, 1/8, 1/8)

`H(p)` average surprise in bits

symbol	probability	codeword
A	1/2	0
B	1/4	10
C	1/8	110
D	1/8	111

Shannon (1948), §§6,9–12; original course examples

EXERCISE

Entropy

Which binary source is hardest to predict?

Each p lists masses p(x) for the same two outcomes x. Compare A: (1,0), B: (3/4,1/4), C: (1/2,1/2).

$$H(p) = -\sum_{x} p(x) \log_2 p(x)$$

H entropy in bits; 0 log 0 = 0

A. A has the largest entropy

B. B has the largest entropy

C. C has the largest entropy

Goodfellow et al., Deep Learning (2016), §3.13

DEFINITION

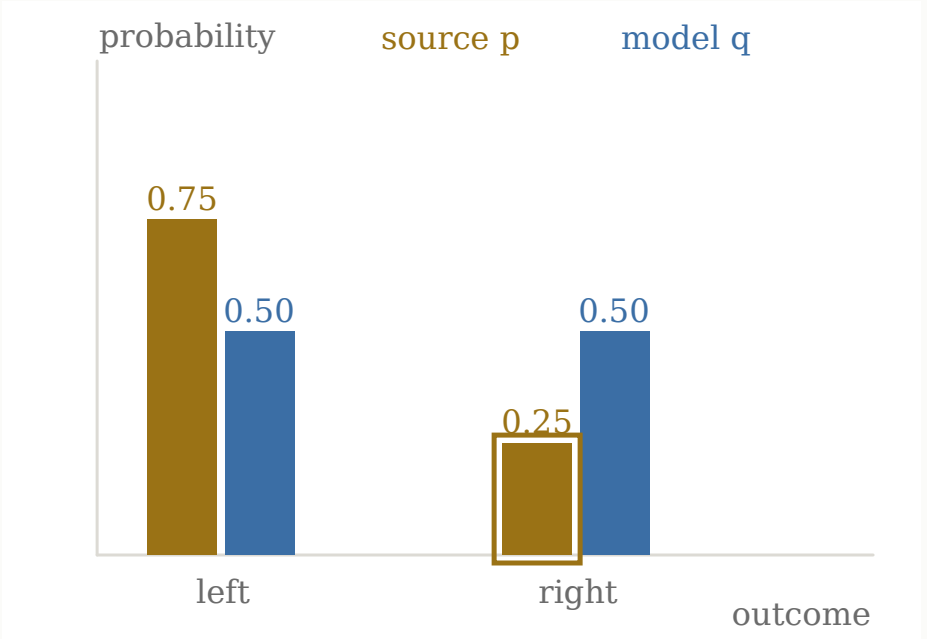
Cross entropy

The source and the prediction can differ

Cross entropy is the expected surprisal under model q when outcomes follow source p. Require q(x)>0 wherever p(x)>0.

$$H(p, q) = -\sum_{x} p(x) \log_2 q(x)$$

- `p` data-generating probabilities
- `q` model probabilities
- `H(p,q)` cross entropy in bits



Goodfellow et al., Deep Learning (2016), §3.13

DEFINITION

KL divergence

The excess cost of using the wrong probabilities

Kullback–Leibler (KL) divergence is the expected log ratio of source p to model q. Use q(x)>0 wherever p(x)>0.

$$D_{\mathrm{KL}}(p\|q) = \sum_{x} p(x) \log_2 \frac{p(x)}{q(x)}$$

`p,q` source and model on the same outcomes

`D_KL` Kullback–Leibler divergence in bits

Goodfellow et al., Deep Learning (2016), §3.13

DERIVATION

KL divergence

The cross-entropy identity follows by expansion

For source p and model q on the same outcomes, assume q(x)>0 wherever p(x)>0. Split the log ratio.

$$H(p, q) = H(p) + D_{\mathrm{KL}}(p \| q)$$

- `H(p)` source entropy
- `D_KL` directed divergence from p to q
- `H(p,q)` cross entropy, in bits

Goodfellow et al., Deep Learning (2016), §3.13

KL divergence

The cross-entropy identity follows by expansion

Expand the log ratio

p is the source and q the model. Sum only over outcomes x with p(x)>0, where q(x)>0.

1. $D_{\mathrm{KL}}(p\|q) = \sum_x p(x) \log_2 [p(x)/q(x)]$.
2. $= \sum_x p(x) \log_2 p(x) - \sum_x p(x) \log_2 q(x)$ by the log-ratio rule.

Goodfellow et al., Deep Learning (2016), §3.13

KL divergence

The cross-entropy identity follows by expansion

Identify the two entropy terms

H(p) is source entropy; H(p,q) is cross entropy.

1. $\sum_x p(x) \log_2 p(x) = -H(p)$.
2. $-\sum_x p(x) \log_2 q(x) = H(p, q)$.
3. Hence $D_{\mathrm{KL}}(p \| q) = H(p, q) - H(p)$, or $H(p, q) = H(p) + D_{\mathrm{KL}}(p \| q)$.

Goodfellow et al., Deep Learning (2016), §3.13

EXAMPLE

KL divergence

A two-route model has a measurable penalty

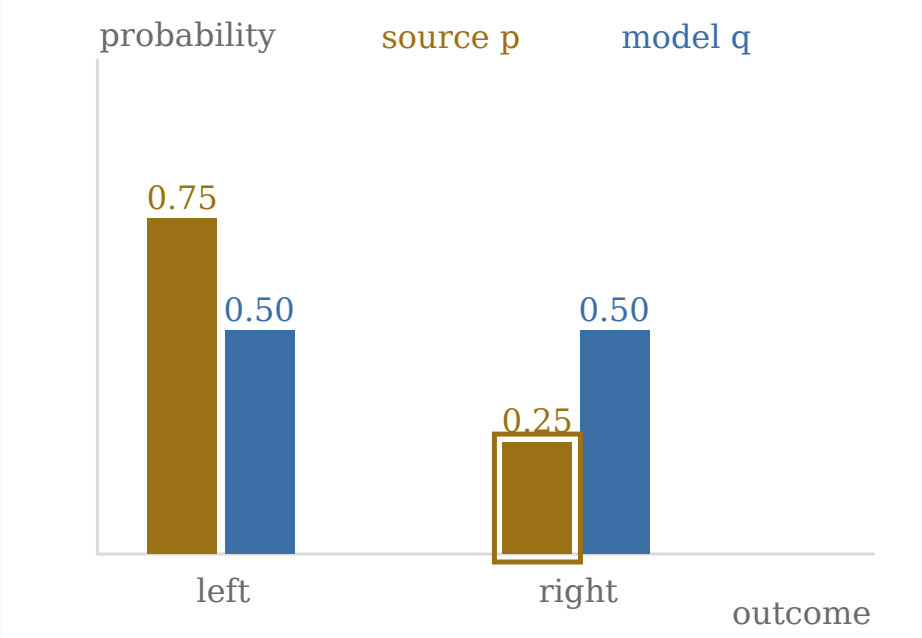
COURSE-CREATED WORKED EXAMPLE

Source p=(3/4,1/4). Model q=(1/2,1/2).

$$H(p,q) = 1, \quad H(p) \approx 0.811, \quad D_{\mathrm{KL}}(p\|q) \approx 0.189$$

`p,q` route probabilities in left/right order

`values` bits per observed route



Goodfellow et al., Deep Learning (2016), §3.13

EXERCISE

KL divergence

What changes when source and model swap?

The source is p=(1/2,1/2) and the model is q=(3/4,1/4), both in left/right order. Find KL in bits.

$$D_{\mathrm{KL}}(p\|q) = \sum_{x} p(x)\log_2[p(x)/q(x)]$$

`p,q` source and model on the same two outcomes

A About 0.189

B About 0.208

C Exactly zero

Goodfellow et al., Deep Learning (2016), §3.13

DERIVATION

Cross entropy

Maximum likelihood uses the same average penalty

For a fixed dataset, maximum likelihood chooses parameters that maximize its model probability. Assume independent discrete observations under the same model qθ.

$$-\frac{1}{n}\log\prod_{i=1}^{n} q_\theta(x_i) = -\frac{1}{n}\sum_{i=1}^{n}\log q_\theta(x_i)$$

- $q_\theta(x_i)$ model mass of observation x_i
- θ adjustable parameters
- n fixed sample count; log is natural

Goodfellow et al., Deep Learning (2016), §5.5

Cross entropy

Maximum likelihood uses the same average penalty

Turn a likelihood product into a sum

n observed values x_i are iid under the model q_θ; $L(\theta)$ is their likelihood; log is natural.

1. $L(\theta) = \prod_{i=1}^{n} q_\theta(x_i)$ by the iid model.
2. $-n^{-1} \log L(\theta) = -n^{-1} \sum_i \log q_\theta(x_i)$ by the log-product rule.

Goodfellow et al., Deep Learning (2016), §5.5

Cross entropy

Maximum likelihood uses the same average penalty

Group repeated outcomes

N_x is the count of outcome x; empirical frequency is p̂ₙ(x)=N_x/n.

1. $-n^{-1}\sum_i \log q_\theta(x_i) = -\sum_x (N_x/n)\log q_\theta(x)$.
2. $= -\sum_x \hat{p}_n(x)\log q_\theta(x)$, the empirical cross entropy.
3. The minus sign converts maximization to minimization; fixed 1/n does not change the optimizer.

Goodfellow et al., Deep Learning (2016), §5.5

EXERCISE

KL divergence

Can a model forbid an event that occurs?

Source p=(0.9,0.1), model q=(1,0). The support of a discrete distribution is its positive-probability outcomes. Is KL(p‖q) finite?

A KL(p||q) is infinite

B KL(p||q) is zero

C The zero-probability term is ignored

Goodfellow et al., Deep Learning (2016), §3.13

DERIVATION

KL divergence

Why the total cannot be negative

For p(x)>0, assume q(x)>0 and set u=q(x)/p(x). Use −ln u≥1−u. Natural logs give KL in nats.

$$D_{\mathrm{KL}}(p\|q) \geq \sum_{p(x)>0} [p(x) - q(x)] \geq 0$$

- `u` q(x)/p(x)
- `log` natural logarithm here; base changes only scale

Goodfellow et al., Deep Learning (2016), §3.13

KL divergence

Why the total cannot be negative

Apply the tangent inequality

Use natural logs; u=q(x)/p(x)>0 on the support of p.

1. $-\log u \geq 1 - u$, with equality only at u=1.
2. $p(x)\log[p(x)/q(x)] \geq p(x) - q(x)$ after multiplying by p(x)>0.

Goodfellow et al., Deep Learning (2016), §3.13

KL divergence

Why the total cannot be negative

Sum over source support

S={x:p(x)>0}; both p and q have total mass 1.

1. $D_{\mathrm{KL}}(p\|q) \geq \sum_{x\in S}(p(x)-q(x))$.
2. $= 1 - \sum_{x\in S} q(x) \geq 0$.
3. Equality requires q=p on S and no q mass outside S. A log-base change preserves the sign.

Goodfellow et al., Deep Learning (2016), §3.13

COUNTEREXAMPLE

KL divergence

Nonnegative does not make it a distance

COURSE-CREATED WORKED EXAMPLE

KL can be asymmetric and violate the triangle inequality.

$$D(p\|r) \approx 1.758$$

$$D(p\|q) + D(q\|r) \approx 0.368 + 0.511 < 1.758$$

`p,q,r` (0.9,0.1), (0.5,0.5), (0.1,0.9)

`D` KL in nats for this example

Goodfellow et al., Deep Learning (2016), §3.13

INTERPRETATION

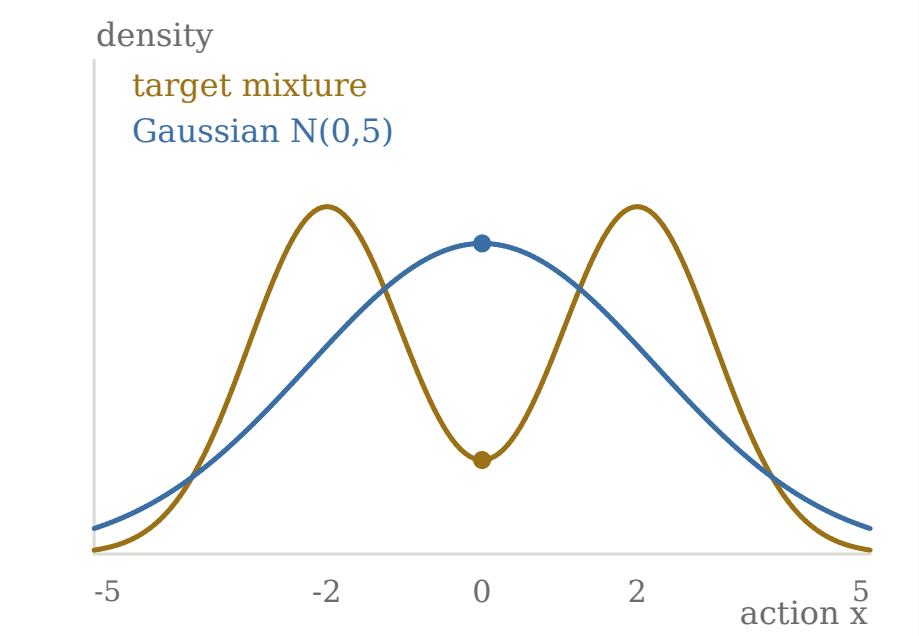
KL fitting

The direction changes what the model averages

A mode maximizes probability mass or density. Fix target p and restrict adjustable qθ to a chosen model family.

$$\min_{\theta} D(p\|q_\theta) \quad \text{or} \quad \min_{\theta} D(q_\theta\|p)$$

- `θ` adjustable parameters
- `p,qθ` target and candidate densities
- `D` KL divergence in the written direction



Goodfellow et al., Deep Learning (2016), §3.13

DERIVATION

KL fitting

Forward KL within a Gaussian family

Continuous density p on R: finite second moment, finite differential entropy, variance v>0. Fit q=N(m,s²), s²>0.

$$-\mathbb{E}_p[\log q(X)] = \tfrac{1}{2}\log(2\pi s^2) + \frac{\mathbb{E}_p[(X-m)^2]}{2s^2}$$

- $\mathbb{E}_p$ expectation under target density p
- X random action from p
- m, s^2 candidate Gaussian mean and variance

Goodfellow et al., Deep Learning (2016), §§3.8,3.9.3,3.13; course derivation

KL fitting

Forward KL within a Gaussian family

Insert the Gaussian model density

p is the fixed target density. q=N(m,s²), with adjustable m and s²>0.

1. $q(x) = (2\pi s^2)^{-1/2} \exp[-(x-m)^2/(2s^2)]$.
2. $-\log q(x) = \frac{1}{2}\log(2\pi s^2) + (x-m)^2/(2s^2)$.

Goodfellow et al., Deep Learning (2016), §§3.8,3.9.3,3.13; course derivation

KL fitting

Forward KL within a Gaussian family

Average under the continuous target

$\mu=\mathrm{E}_p[X]$, $v=\mathrm{Var}_p(X)>0$; differential entropy $h(p)=-\int p(x)\log p(x)\,dx$ is finite.

1. $C(m, s^2) = -\mathbb{E}_p[\log q(X)] = \frac{1}{2}\log(2\pi s^2) + \mathbb{E}_p[(X-m)^2]/(2s^2).$
2. $D_{\mathrm{KL}}(p\|q) = \int p(x)\log[p(x)/q(x)]\,dx = C(m, s^2) - h(p).$
3. Finite fixed h(p) makes minimizing KL equivalent to minimizing C.

Goodfellow et al., Deep Learning (2016), §§3.8,3.9.3,3.13; course derivation

DERIVATION

KL fitting

The best Gaussian center is the target mean

Target mean $\mu = E_p[X]$, variance $v = E_p[(X-\mu)^2]$. Adjustable m is the Gaussian mean; its variance $s^2 > 0$ stays fixed.

$$\mathbb{E}_p[(X-m)^2] = v + (\mu - m)^2$$

- E_p expectation for X drawn from target p
- v target variance
- m candidate Gaussian mean

Goodfellow et al., Deep Learning (2016), §§3.8,3.9.3,3.13; course derivation

KL fitting

The best Gaussian center is the target mean

Expand the square around the target mean

$\mu = E_p[X]$, $v = \mathrm{Var}_p(X)$, m is the adjustable Gaussian mean.

1. $X - m = (X - \mu) + (\mu - m).$
2. $(X - m)^2 = (X - \mu)^2 + 2(X - \mu)(\mu - m) + (\mu - m)^2.$

Goodfellow et al., Deep Learning (2016), §§3.8,3.9.3,3.13; course derivation

KL fitting

The best Gaussian center is the target mean

Average and minimize

μ−m is fixed when taking expectation over X.

1. $\mathbb{E}[X-\mu]=0$, so the cross term vanishes.
2. $\mathbb{E}[(X-m)^2]=v+(\mu-m)^2$.
3. For fixed $s^2>0$, the nonnegative square is minimized at m=μ.

Goodfellow et al., Deep Learning (2016), §§3.8,3.9.3,3.13; course derivation

DERIVATION

KL fitting

The best Gaussian variance matches the spread

Fix m=μ. Target variance v>0 is fixed. Adjustable variance r=s²>0 minimizes the expected negative log density C(r).

$$C(r) = \tfrac{1}{2}\log(2\pi r) + \frac{v}{2r}, \qquad C\prime(r) = \frac{r - v}{2r^2}$$

- r candidate Gaussian variance
- v positive target variance

Goodfellow et al., Deep Learning (2016), §§3.8,3.9.3,3.13; course derivation

KL fitting

The best Gaussian variance matches the spread

Differentiate with respect to the variance

r=s²>0 is adjustable; v>0 is the target variance.

1. $C(r) = \frac{1}{2}\log(2\pi r) + v/(2r).$
2. $C\prime(r) = 1/(2r) - v/(2r^2) = (r - v)/(2r^2).$

Goodfellow et al., Deep Learning (2016), §§3.8,3.9.3,3.13; course derivation

KL fitting

The best Gaussian variance matches the spread

Verify the minimum, not just stationarity

The denominator $2r^2$ is positive for $r>0$.

1. For $r<v$, $C'(r)<0$. For $r>v$, $C'(r)>0$.
2. C decreases toward v and increases after v, so $r=v$ is the global minimum.
3. The optimal Gaussian has mean μ and variance v.

Goodfellow et al., Deep Learning (2016), §§3.8,3.9.3,3.13; course derivation

EXAMPLE

KL fitting

One Gaussian can place its mean between two peaks

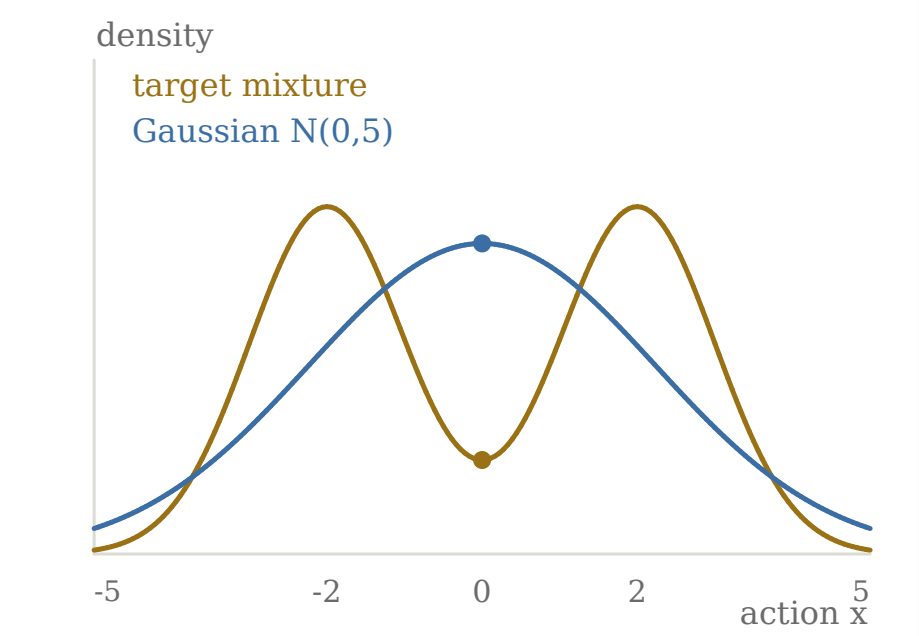
COURSE-CREATED WORKED EXAMPLE

A mixture first chooses a component, then samples it. Here each component $\mathrm{N}(-2,1)$ or $\mathrm{N}(2,1)$ has weight $1/2$ and variance 1.

$$\mu = 0, \qquad v = 1 + 4 = 5, \qquad q^* = \mathcal{N}(0, 5)$$

`μ,v` mean and variance of the target mixture

`q*` best forward-KL Gaussian, with mean μ and variance v



Goodfellow et al., Deep Learning (2016), §§3.8,3.9.3,3.13; course derivation

EXERCISE

KL fitting

An uneven mixture shifts both fitted moments

Target p=0.75 N(−2,1)+0.25 N(2,1). Each component has variance 1. Fit one Gaussian by forward KL.

$$m = \mathbb{E}_p[X], \qquad s^2 = \mathbb{E}_p[X^2] - m^2$$

`m,s²` forward-KL Gaussian mean and variance

A. Mean −1, variance 1

B. Mean −1, variance 4

C. Mean 0, variance 5

Goodfellow et al., Deep Learning (2016), §§3.8,3.9.3,3.13; course derivation

INTERPRETATION

KL fitting

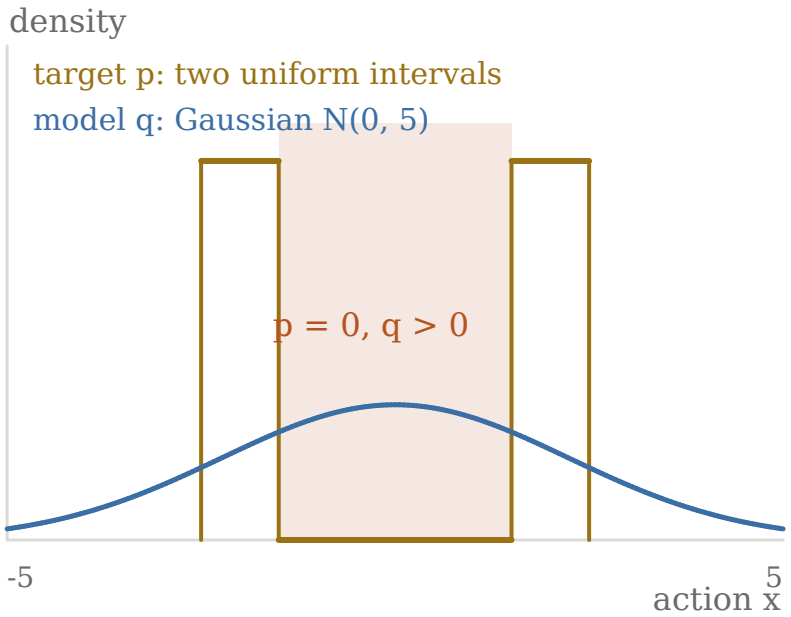
Reverse KL penalizes probability in a forbidden gap

If a region has zero target density p but positive model probability under q, reverse KL is infinite.

$$D(q\|p) = \mathbb{E}_q[\log q(X) - \log p(X)]$$

`p,q` target and candidate densities

`X` action sampled from q



Goodfellow et al., Deep Learning (2016), §3.13

EXERCISE

KL fitting

Does reverse KL always select one mode?

A bimodal target p has two peaks. The candidate family for q contains p itself. Compare the minimum forward and reverse KL.

A. Yes, one mode is always optimal

B. No, q=p minimizes both directions

C. Neither direction can reach zero

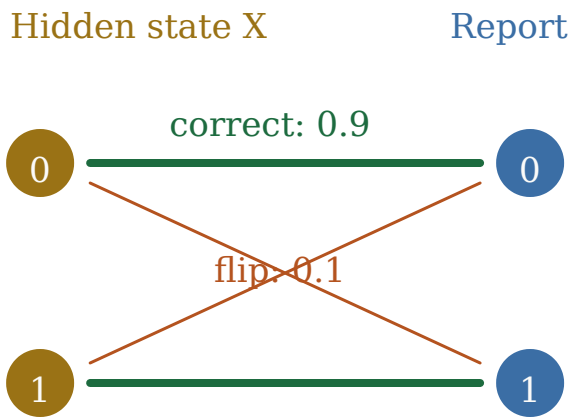
Goodfellow et al., Deep Learning (2016), §3.13

MOTIVATION

Mutual information

A sensor is useful when it changes our uncertainty

Let X be a hidden state and Y a sensor report. Does knowing Y help us predict X?



$P(X=0) = P(X=1) = 0.5$

Weissman, Stanford EE376A (2018), Lecture3, Definitions5–6, eqs.(30)–(46)

EXAMPLE

Mutual information

The marginals come from summing the joint table

COURSE-CREATED WORKED EXAMPLE

X is a fair binary state. For either state, the binary report Y equals X with probability 0.9.

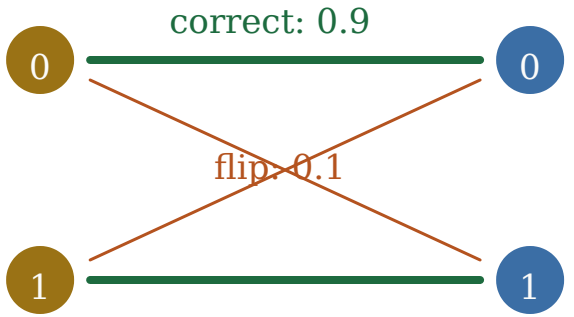
$$p_{XY} = \begin{bmatrix} 0.45 & 0.05 \\ 0.05 & 0.45 \end{bmatrix}$$

`p_XY` joint mass table for X,Y

`rows` X=0,1

`columns` Y=0,1

`p_X,p_Y` marginals, each (1/2,1/2)



P(X=0) = P(X=1) = 0.5

Weissman, Stanford EE376A (2018), Lecture3, Definitions5–6, eqs.(30)–(46)

DEFINITION

Mutual information

Compare dependence with the independent baseline

For discrete X,Y, mutual information compares the joint mass to the product of the marginals by KL. Sum over pairs with positive joint mass.

$$I(X;Y) = \sum_{x,y} p(x,y) \log_2 \frac{p(x,y)}{p_X(x) p_Y(y)}$$

`I(X;Y)`	mutual information in bits
`x,y`	values of X,Y
`p(x,y)`	joint mass at (x,y)
`p_X,p_Y`	marginal mass functions

Weissman, Stanford EE376A (2018), Lecture3, Definitions5–6, eqs.(30)–(46)

DERIVATION

Mutual information

Information is expected uncertainty reduction

Conditional entropy H(X|Y) averages uncertainty about state X after observing report Y. Factor joint mass into report mass and conditional state mass.

$$I(X;Y) = H(X) - H(X \mid Y)$$

`I(X;Y)` mutual information in bits

`H(X)` state entropy in bits

`H(X|Y)` report-weighted state entropy

Weissman, Stanford EE376A (2018), Lecture3, Definitions5–6, eqs.(30)–(46)

Mutual information

Information is expected uncertainty reduction

Cancel the report probability

For p(x,y)>0, factor the joint mass into report mass p_Y(y) and conditional state mass p(x|y).

1. $\log_2 \frac{p(x,y)}{p_X(x)p_Y(y)} = \log_2 \frac{p(x|y)}{p_X(x)}.$
2. $= \log_2 p(x \mid y) - \log_2 p_X(x).$

Weissman, Stanford EE376A (2018), Lecture3, Definitions5–6, eqs.(30)–(46)

Mutual information

Information is expected uncertainty reduction

Average over the joint distribution

H(X|Y)=−Σₓ,ᵧp(x,y)log₂p(x|y) is conditional entropy.

1. $I(X;Y) = \sum_{x,y} p(x,y) \log_2 p(x \mid y) - \sum_{x,y} p(x,y) \log_2 p_X(x)$.
2. The first sum is −H(X|Y). Summing over y makes the second sum −H(X).
3. Therefore I(X;Y)=H(X)−H(X|Y), an expected reduction.

Weissman, Stanford EE376A (2018), Lecture3, Definitions5–6, eqs.(30)–(46)

EXAMPLE

Mutual information

A noisy binary sensor supplies part of a bit

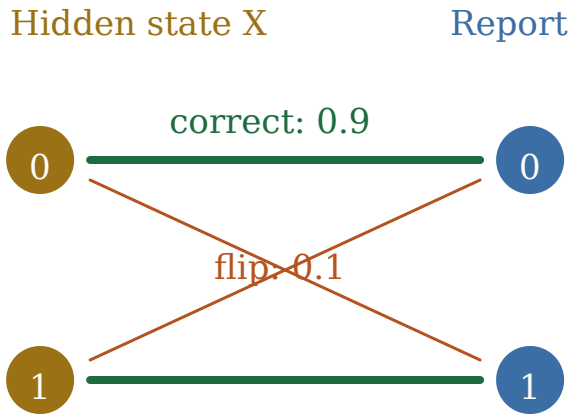
COURSE-CREATED WORKED EXAMPLE

X is fair; Y flips it independently with probability e=0.1. Binary entropy $h_2(e)$ is the entropy of probabilities (e,1−e).

$$I(X;Y) = 1 - h_2(e) \approx 0.531$$

`h_2(e)` $-e\log_2 e-(1-e)\log_2(1-e)$

`e` independent binary flip probability



P(X=0) = P(X=1) = 0.5

Weissman, Stanford EE376A (2018), Lecture3, Definitions5–6, eqs.(30)–(46)

EXERCISE

Mutual information

A noisier sensor leaves more uncertainty

X is a fair binary state. Report Y flips X independently with probability e=0.25. Find the mutual information I(X;Y).

$$I(X;Y) = H(X) - H(X \mid Y)$$

`H(X)` state entropy

`H(X|Y)` average remaining state entropy

A About 0.811 bits

B About 0.189 bits

C Exactly 0.75 bits

Weissman, Stanford EE376A (2018), Lecture3, Definitions5–6, eqs.(30)–(46)

EXAMPLE

Mutual information

Zero covariance can hide a nonlinear dependence

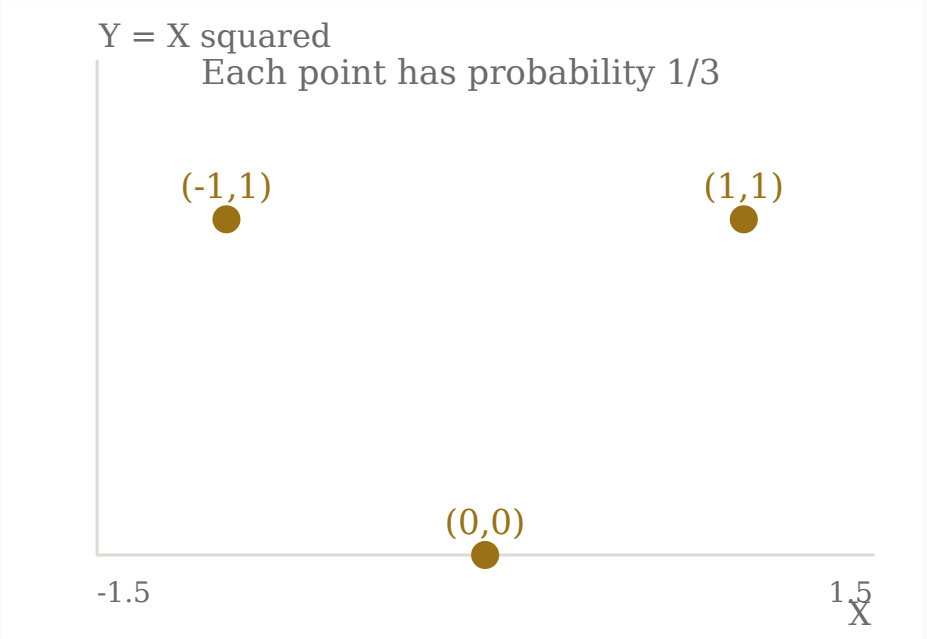
COURSE-CREATED WORKED EXAMPLE

X is uniform on $\{-1,0,1\}$. Set $Y=X^2$.

$$\operatorname{Cov}(X,Y) = \mathbb{E}[X^3] - \mathbb{E}[X]\mathbb{E}[X^2] = 0$$

`X,Y` random input and its square

`Cov` covariance; both variances are positive



[Goodfellow et al., Deep Learning (2016), §3.8; original example]

EXERCISE

Mutual information

How much does the square reveal?

X is uniform on $\{-1,0,1\}$; $Y=X^2$. $H(Y|X)=0$.

$$I(X;Y) = H(Y) - H(Y \mid X)$$

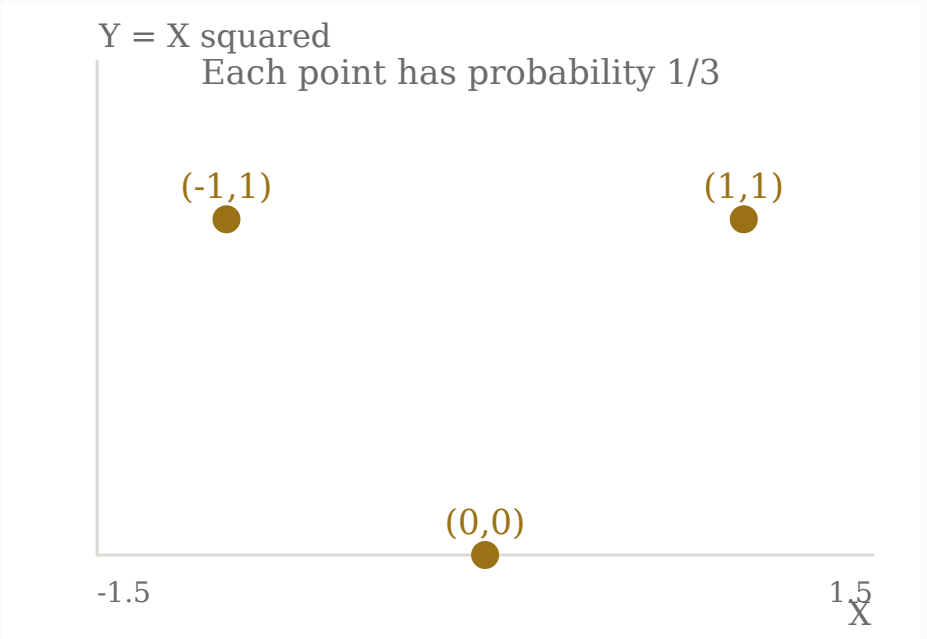
`H(Y)` entropy of the squared output

`h₂(a)` entropy of binary masses (a,1−a)

A. I(X;Y)=0 because covariance is zero

B. $I(X;Y)=\log_2 3$

C. $I(X;Y)=h_2(1/3)\approx 0.918$ bits



Weissman, Stanford EE376A (2018), Lecture3, Definitions5–6, eqs.(30)–(46)

EXERCISE

Mutual information

Compare a direct and an inverted sensor

X is a fair binary state. Sensor A gives $Y=X$. Sensor B gives $Y=1-X$.

A. A has more mutual information

B. Both provide 1 bit

C. B provides zero information

Weissman, Stanford EE376A (2018), Lecture3, Definitions5–6, eqs.(30)–(46)

RECAP

From distributions to data

The formulas need evidence from observations

Entropy measures source uncertainty. KL measures model mismatch. Mutual information measures dependence. Data let us estimate expectations.

Route choices
Uncertainty within the source

Model predictions
Mismatch from the source

Sensor reports
Dependence with hidden state

Trial data
Estimate unknown quantities

Goodfellow et al., Deep Learning (2016), §3.13

DEFINITION

Monte Carlo estimation

An average estimates a chosen expectation

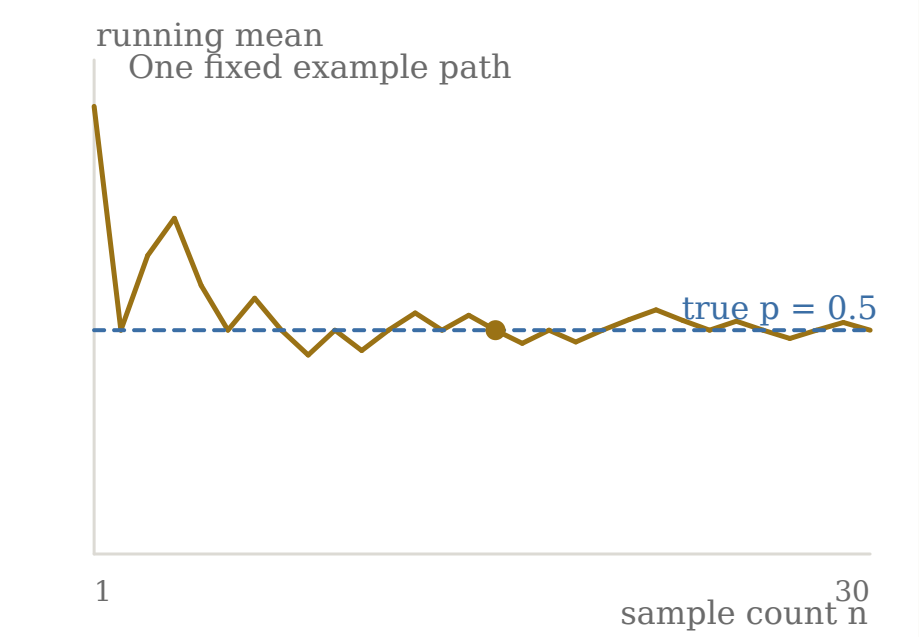
Monte Carlo estimation approximates an expectation by averaging sampled values. The count n is fixed and the draws are independent and identically distributed (iid).

$$\hat{\mu}_n = \frac{1}{n}\sum_{i=1}^{n} g(X_i), \qquad \mu = \mathbb{E}_p[g(X)]$$

`X_i` draws from source p

`g,mu` quantity measured and its true mean

`μ̂ₙ` average of n measured values



UC Berkeley CS70 (2017), Note18: mean, variance, concentration

DERIVATION

Monte Carlo estimation

Linearity explains unbiasedness

An estimator is unbiased if its expectation equals its target. Here n is fixed and each g(Xᵢ) has finite expectation μ.

$$\mathbb{E}[\hat{\mu}_n] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[g(X_i)] = \mu$$

- $\hat{\mu}$ random average of n measurements
- $g(X_i)$ measurement from draw i
- μ common mean

UC Berkeley CS70 (2017), Note18: mean, variance, concentration

Monte Carlo estimation

Linearity explains unbiasedness

Use linearity of expectation

n is fixed; every measured value g(Xᵢ) has the same finite expectation μ.

1. $\mathbb{E}[\hat{\mu}_n] = \mathbb{E}[n^{-1} \sum_i g(X_i)]$.
2. $= n^{-1} \sum_i \mathbb{E}[g(X_i)]$ by linearity.
3. $= n^{-1}(n\mu) = \mu$ by the common sample expectation.

[UC Berkeley CS70 (2017), Note18: mean, variance, concentration]

DERIVATION

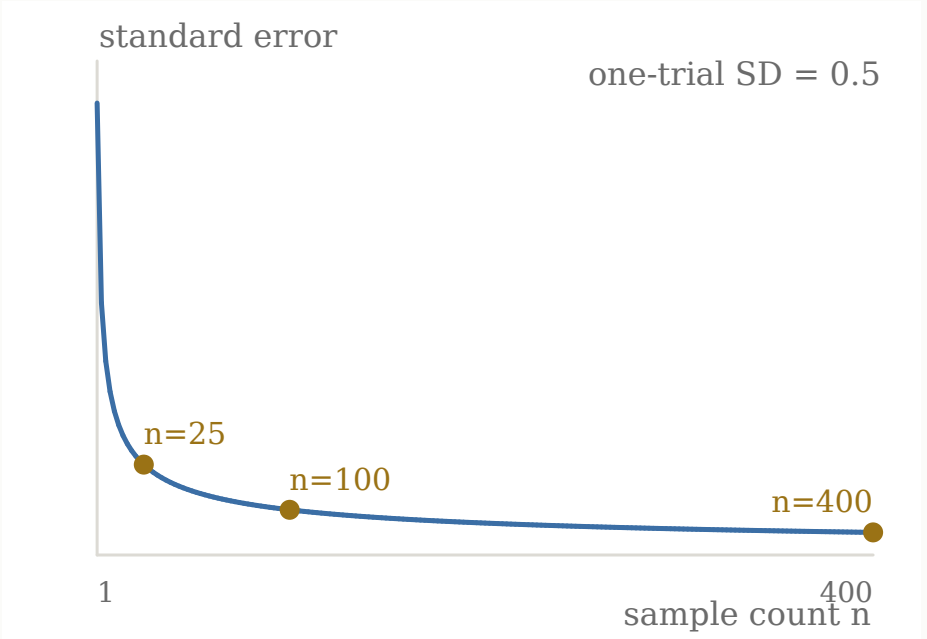
Monte Carlo estimation

Independence makes the variance shrink

Standard error (SE) is an estimator’s standard deviation. Use n fixed independent measurements with common finite variance σ².

$$\mathrm{Var}(\hat{\mu}_n) = \frac{\sigma^2}{n}, \qquad \mathrm{SE}(\hat{\mu}_n) = \frac{\sigma}{\sqrt{n}}$$

- $\hat{\mu}_n$ mean of n measurements
- σ, σ^2 one measurement’s standard deviation and variance
- SE standard deviation of the mean



[UC Berkeley CS70 (2017), Note18: mean, variance, concentration]

Monte Carlo estimation

Independence makes the variance shrink

Expand the variance of the average

$\hat{\mu}_n$ averages n independent measurements Z_i with common finite variance σ^2.

1. $\operatorname{Var}(\hat{\mu}_n) = n^{-2} \operatorname{Var}(\sum_i Z_i)$.
2. $= n^{-2} \sum_i \operatorname{Var}(Z_i)$ because all cross covariances vanish.
3. $= n^{-2}(n\sigma^2) = \sigma^2/n$.

UC Berkeley CS70 (2017), Note18: mean, variance, concentration

Monte Carlo estimation

Independence makes the variance shrink

Convert variance to standard error

σ is the standard deviation of one measurement. SE is the standard deviation of the mean $\hat{\mu}_n$.

1. $\mathrm{SE}(\hat{\mu}_n) = \sqrt{\mathrm{Var}(\hat{\mu}_n)} = \sigma/\sqrt{n}$.
2. Four times as many independent samples gives half the standard error.
3. This describes the average, not the spread of one observation.

UC Berkeley CS70 (2017), Note18: mean, variance, concentration

EXERCISE

Monte Carlo estimation

What does four times as much data buy?

Independent draws have the same finite variance σ^2. Increase the fixed count n from 100 to 400. What happens to the standard error of the mean?

$$\mathrm{SE}(\hat{\mu}_n) = \sigma / \sqrt{n}$$

σ unchanged one-trial standard deviation

A Standard error becomes one quarter

B Standard error becomes one half

C Every realized estimate is closer

UC Berkeley CS70 (2017), Note18: mean, variance, concentration

DERIVATION

Monte Carlo estimation

A variance bound gives a probability statement

Chebyshev’s inequality bounds deviation probability using variance. Here $\hat{\mu}_n$ is an unbiased iid average; n is fixed and $\varepsilon > 0$.

$$P(|\hat{\mu}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2}$$

- $\hat{\mu}_n$, μ — sample mean and true mean
- epsilon — chosen error tolerance
- σ^2 — finite variance per measurement

UC Berkeley CS70 (2017), Note18: mean, variance, concentration

Monte Carlo estimation

A variance bound gives a probability statement

Bound squared error on the bad event

$E=|\hat{\mu}_n-\mu|$ is the absolute error of the sample mean $\hat{\mu}_n$ around the true mean μ. $A=\{E\geq\varepsilon\}$, with $\varepsilon>0$.

1. On A, $E^2\geq\varepsilon^2$; outside A, $E^2\geq 0$.
2. Therefore $\mathbb{E}[E^2] \geq \varepsilon^2 P(A)$.

UC Berkeley CS70 (2017), Note18: mean, variance, concentration

Monte Carlo estimation

A variance bound gives a probability statement

Insert the estimator variance

The fixed-n iid average is unbiased and has variance σ²/n.

1. $\mathbb{E}[E^2] = \mathrm{Var}(\hat{\mu}_n) = \sigma^2/n.$
2. Dividing gives $P(|\hat{\mu}_n - \mu| \geq \varepsilon) \leq \sigma^2/(n\varepsilon^2).$
3. The result is an upper bound, not an equality or a guarantee for each dataset.

UC Berkeley CS70 (2017), Note18: mean, variance, concentration

EXAMPLE

Monte Carlo estimation

A conservative trial count for a success rate

COURSE-CREATED WORKED EXAMPLE

For n fixed independent Bernoulli trials, $p(1-p) \le 1/4$. Require $P(|\hat{p}-p| \ge 0.1) \le 0.05$.

$$\frac{1}{4n(0.1)^2} \le 0.05 \quad \Longrightarrow \quad n \ge 500$$

- $p, \hat{p}$ success probability and observed fraction
- n fixed number of independent trials

UC Berkeley CS70 (2017), Note18: mean, variance, concentration

EXERCISE

Monte Carlo estimation

A tighter tolerance changes sample cost

Use n fixed independent Bernoulli trials and p(1−p)≤1/4. Require failure probability at most 0.05 with error tolerance ε=0.05.

$$P(|\hat{p} - p| \geq \varepsilon) \leq 1/(4n\varepsilon^2)$$

`p-hat,p` sample fraction and true success probability

A 250 trials suffice by this bound

B 1000 trials suffice by this bound

C 2000 trials suffice by this bound

[UC Berkeley CS70 (2017), Note18: mean, variance, concentration]

RESULT

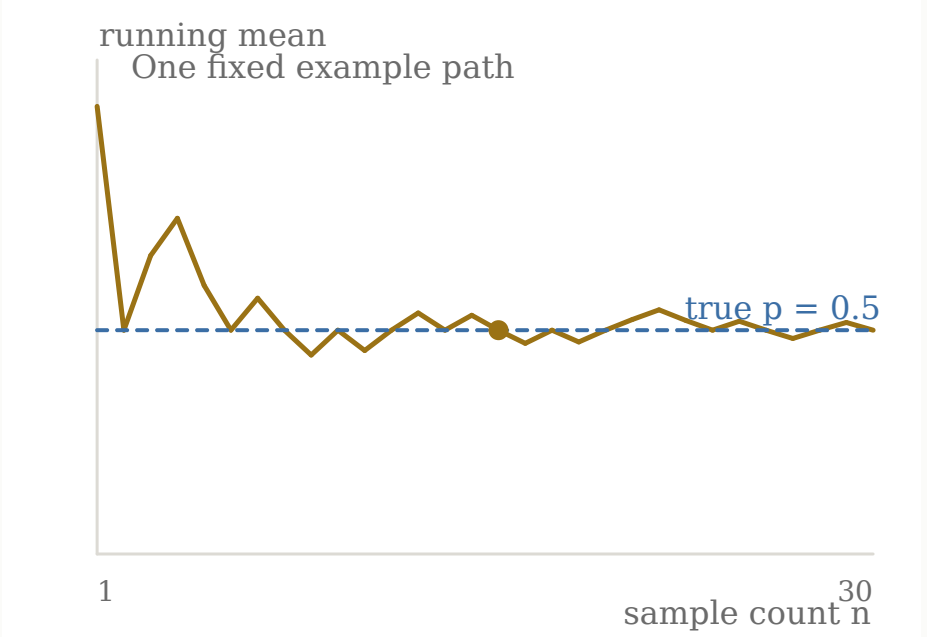
Monte Carlo estimation

The law of large numbers is not a monotone path

Convergence in probability means every fixed positive error tolerance is exceeded with probability tending to zero. Here samples are iid with finite variance.

$$P(|\hat{\mu}_n - \mu| \geq \varepsilon) \longrightarrow 0$$

- $\hat{\mu}_n$ average of n iid measurements
- μ population mean
- ε fixed positive error tolerance



UC Berkeley CS70 (2017), Note18: mean, variance, concentration

COUNTEREXAMPLE

Monte Carlo estimation

Correlated repetitions are not fresh evidence

Let $\bar{Z}$ be the mean of n measurements Z_i with finite variances. Dependence leaves pairwise covariance terms in its variance.

$$\operatorname{Var}(\bar{Z}) = \frac{1}{n^2}\left[\sum_i \operatorname{Var}(Z_i) + 2\sum_{i<j} \operatorname{Cov}(Z_i, Z_j)\right]$$

`Z_i` measured values g(X_i)

`Z-bar` sample average

UC Berkeley CS70 (2017), Note18: mean, variance, concentration

DEFINITION

Stopping rules

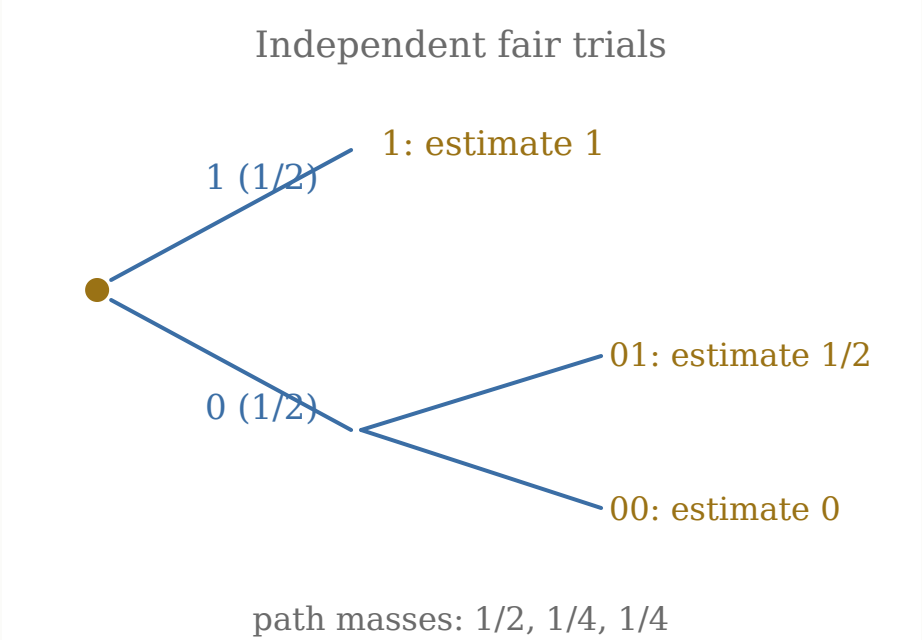
The data can determine when collection ends

A stopping time T is a random end time: whether $T \le t$ can be decided from observations $X_1,\dots,X_t$ alone.

$$T = \begin{cases} 1, & X_1 = 1 \\ 2, & X_1 = 0 \end{cases}$$

`X_i` independent fair Bernoulli trials

`T` random sample count chosen by the rule



Sheffield, MIT 18.175 (2016), Lecture 14, pp.34–35; original example

EXAMPLE

Stopping rules

Enumerating paths exposes the selection effect

COURSE-CREATED WORKED EXAMPLE

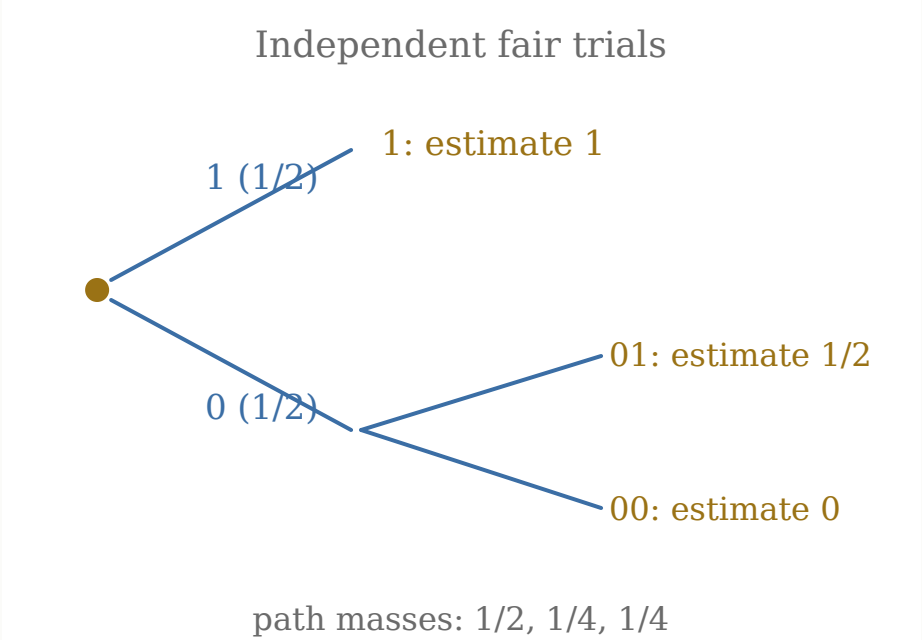
Stop after first success, but take at most two trials. The trials are independent, with success probability 1/2 each.

$$(T, \hat{p}_T) = (1, 1), (2, 1/2), (2, 0)$$

`T` number of trials taken

`p̂T` reported success fraction

`path masses` 1/2, 1/4, 1/4 in order



Sheffield, MIT 18.175 (2016), Lecture 14, pp.34–35; original example

EXERCISE

Stopping rules

Is the stopped success fraction unbiased?

Fair trials: p=1/2. Completed paths 1,01,00 have probabilities 1/2,1/4,1/4 and success fractions 1,1/2,0.

A Its expectation is 1/2

B Its expectation is 5/8

C Its expectation is 1

Sheffield, MIT 18.175 (2016), Lecture 14, pp.34–35; original example

EXERCISE

Stopping rules

Reverse the stopping preference

Trials are independent Bernoulli(1/2). Stop after the first failure or after two trials. Find the expected reported success fraction.

A. Expected estimate 3/8

B. Expected estimate 1/2

C. Expected estimate 5/8

Sheffield, MIT 18.175 (2016), Lecture 14, pp.34–35; original example

APPLICATION

Stopping rules

Match the guarantee to the collection rule

Fixed-n error guarantees assume a sample size fixed before inspection. For data-dependent stopping, use a guarantee proved for the collection rule.

Howard et al. (2020), Introduction: time-uniform bounds

EXAMPLE

Monte Carlo estimation

A rollout result needs an uncertainty statement

COURSE-CREATED WORKED EXAMPLE

A fixed Bernoulli evaluation has 80 successes in 100 independent trials. Approximate 95% interval: [0.7216, 0.8784].

$$\hat{p} = 0.8, \qquad \widehat{\mathrm{SE}} = \sqrt{\frac{0.8(0.2)}{100}} = 0.04$$

`p̂` observed success fraction

`estimated SE` plug-in standard error of the success fraction

NIST Dataplot, Proportion Confidence Limits, Normal approximation

EXERCISE

Monte Carlo estimation

How many draws to see a rare route at least once?

A route has probability 0.1 in each independent draw. Find the smallest fixed integer n with detection probability at least 0.95.

$$P(\text{no rare route}) = (0.9)^n$$

n fixed independent sample count

A 10 draws

B 29 draws

C 95 draws

UC Berkeley CS70 (2017), Note18: mean, variance, concentration

MOTIVATION

Density and modes

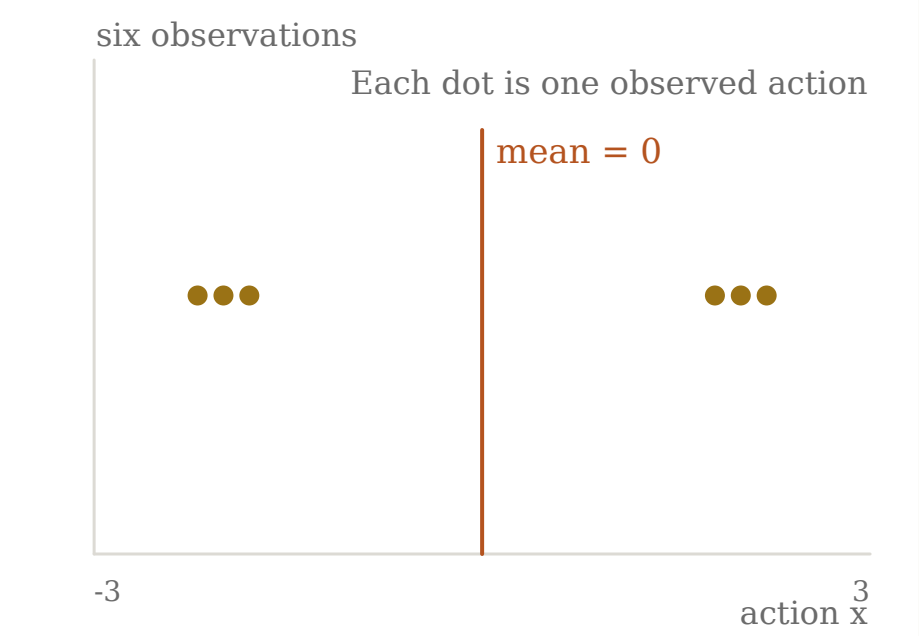
A mean can hide the choices a robot actually makes

Observed actions: -2.2, -2.0, -1.8, 1.8, 2.0, 2.2. Their mean is zero.

$$\bar{x} = 0, \qquad \text{dense groups lie near } -2 \text{ and } 2$$

`x_i` six observed scalar actions

`x̄` arithmetic mean



Wasserman, 36-708 Density Estimation, §§1,3–4

DEFINITION

Density and modes

A discrete mode maximizes probability mass

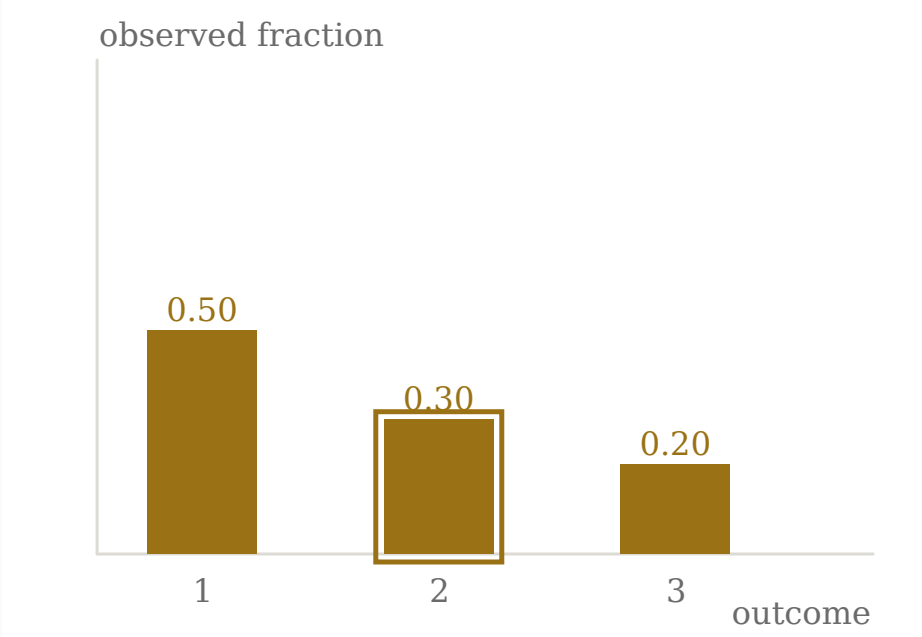
A discrete mode maximizes probability mass under the source. An empirical mode maximizes observed frequency. The figure shows counts 5, 3, 2 out of 10 as proportions.

$$\hat{p}_n(x) = \frac{N_x}{n}, \qquad \hat{m} \in \arg\max_x \hat{p}_n(x)$$

`N_x,n` count in category x and total count

`p̂ₙ(x)` observed fraction in category x

`m̂` any empirical mode



Wasserman, 36-708 Density Estimation, §§1,3–4

EXERCISE

Density and modes

Does the largest observed count prove the true mode?

Three categories have counts 5,3,2 in ten independent observations.

A. The first category is the empirical mode

B. The first category is certainly the population mode

C. The sample mean defines the mode

Wasserman, 36-708 Density Estimation, §§1,3–4

RECALL

Density and modes

Continuous data rarely repeat exactly

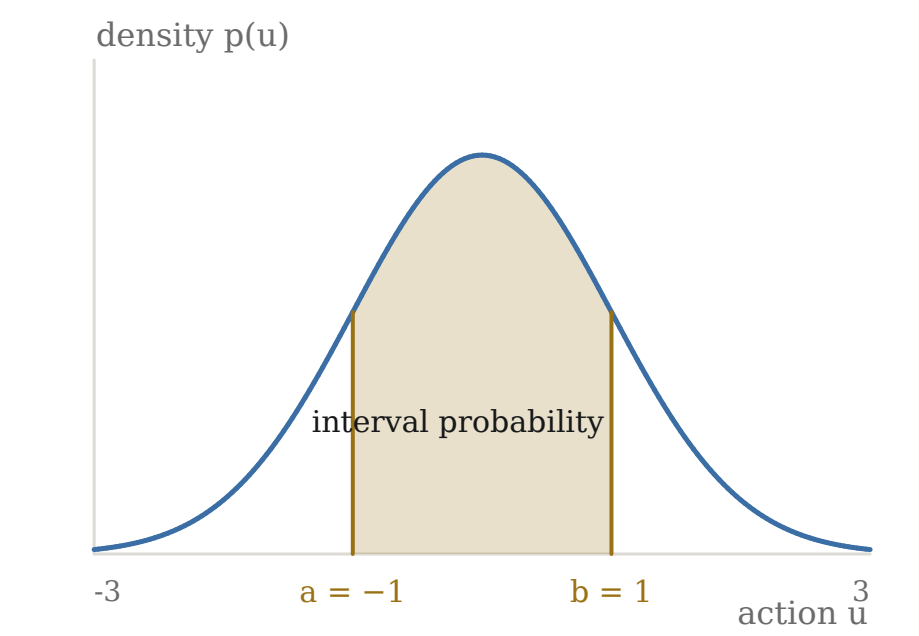
For a model with density p≥0 and ∫p=1, each exact value has probability zero. Interval probability is the area under p over that interval.

$$P(X = x) = 0, \qquad P(a \le X \le b) = \int_a^b p(u)\,du$$

`X,x` random action and a fixed value

`p(u)` density at integration variable u

`a,b` interval endpoints



Goodfellow et al., Deep Learning (2016), §3.3.2

DEFINITION

Histograms

Counts become density after dividing by width

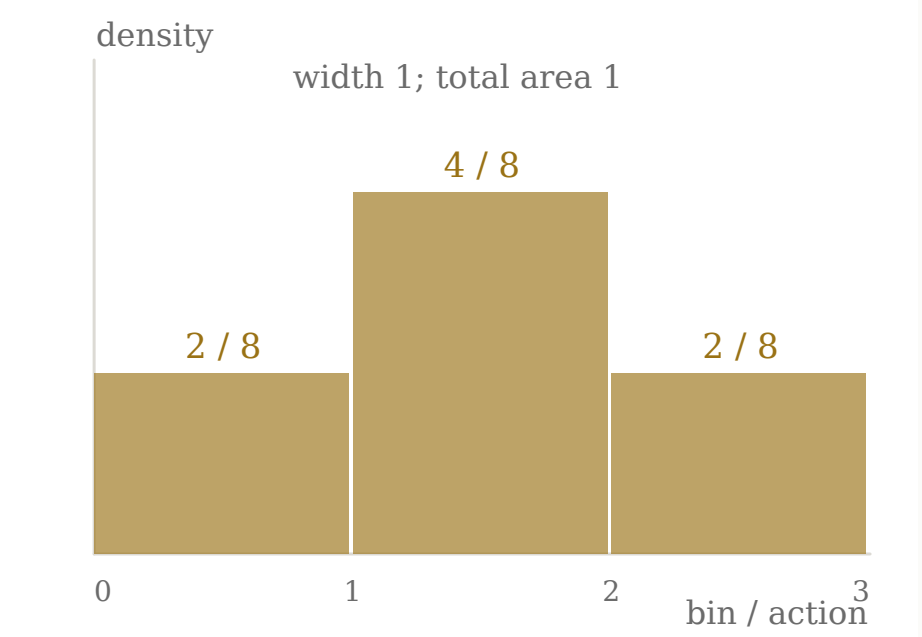
A histogram estimates density as a constant on each bin. Here bins have width h and contain counts N_j out of n observations.

$$\hat{p}(x) = \frac{N_j}{nh} \quad (x \text{ in bin } j)$$

`n,N_j` total count and count in bin j

`h` bin width in action units

`p̂(x)` estimated density at action x



Wasserman, 36-708 Density Estimation, §3; original example

EXAMPLE

Histograms

The area check catches normalization mistakes

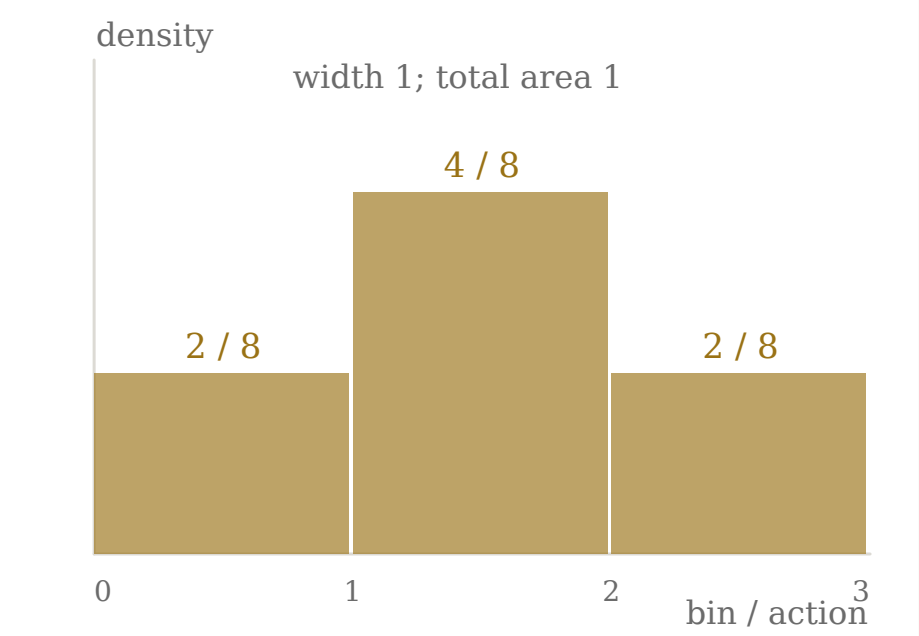
COURSE-CREATED WORKED EXAMPLE

Eight samples give counts 2, 4, 2 in bins [0,1), [1,2), [2,3), each of width 1.

$$\hat{p} = (0.25, 0.5, 0.25), \qquad \sum_j h\hat{p}_j = 1$$

`h` common bin width 1

`p-hat_j` constant density in bin j



Wasserman, 36-708 Density Estimation, §3; original example

EXERCISE

Histograms

Compare bins with different widths

n=7. Bin A: count 4,width 2. Bin B: count 3,width 1. Which has higher histogram density?

$$\hat{p}_j = N_j / (n h_j)$$

`h_j` width of bin j; N_j is its count

A Bin A

B Bin B

C They have equal density

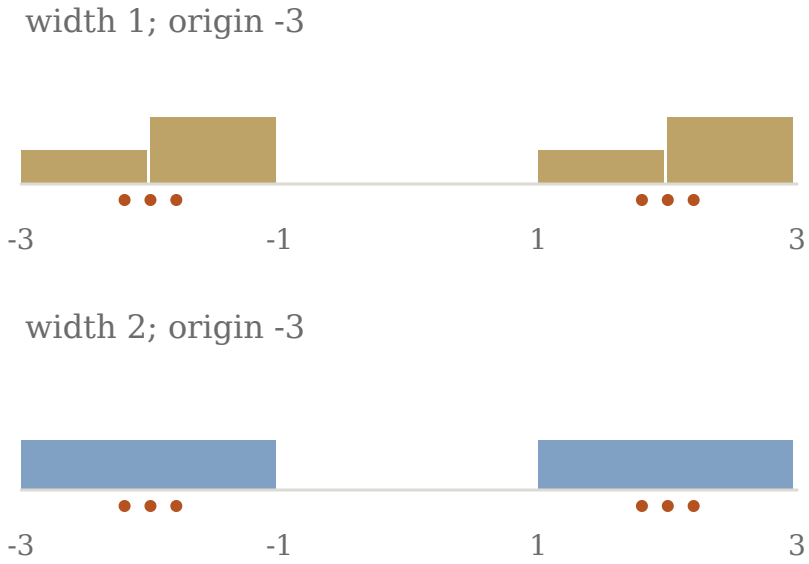
Wasserman, 36-708 Density Estimation, §3; original example

INTERPRETATION

Histograms

The apparent peak depends on the binning

Use observations $-2.2, -2, -1.8, 1.8, 2, 2.2$. Compare half-open bins of width 1 and 2, both starting at -3.



Wasserman, 36-708 Density Estimation, §3; original example

DEFINITION

Kernel density estimation

Each sample contributes a small smooth bump

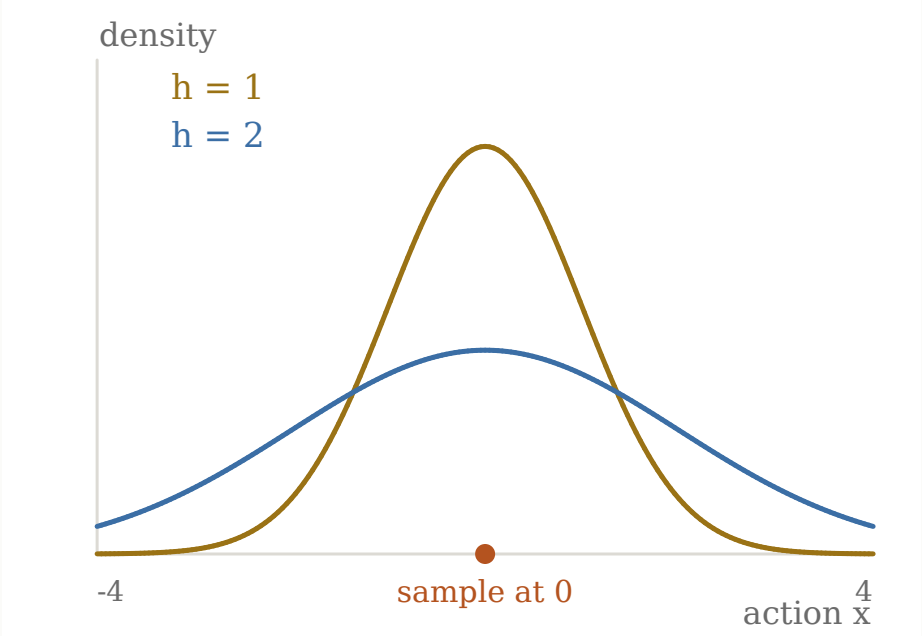
Kernel density estimation (KDE) averages one rescaled bump per sample. Use a nonnegative kernel K with integral 1 and bandwidth h>0.

$$\hat{p}_h(x) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{x - x_i}{h} \right)$$

`x_i,x` sample i and query location

`n` sample count

`h,K` bandwidth and unit-integral kernel



[Wasserman, 36-708 Density Estimation, §4; original example]

DERIVATION

Kernel density estimation

The normalization survives a change of scale

Kernel K has area 1 over ℝ. Substitute u=(x−xᵢ)/h, so dx=h du.

$$\int_{\mathbb{R}} \frac{1}{h} K\left(\frac{x - x_i}{h}\right) dx = \int_{\mathbb{R}} K(u)\, du = 1$$

- `h` positive bandwidth
- `x_i` fixed sample center
- `u` rescaled integration variable

Wasserman, 36-708 Density Estimation, §4; original example

Kernel density estimation

The normalization survives a change of scale

Change the integration variable

K has integral 1 on $\mathbb{R}$, h>0, and x_i is a fixed sample center.

1. Set $u = (x - x_i)/h$, so $dx = h\,du$.
2. $\int_{\mathbb{R}} h^{-1} K((x - x_i)/h)\,dx = \int_{\mathbb{R}} h^{-1} K(u) h\,du = 1.$

Wasserman, 36-708 Density Estimation, §4; original example

Kernel density estimation

The normalization survives a change of scale

Average the unit-area bumps

The estimated density $\hat{p}_h(x)$ averages n unit-area bumps; x spans $\mathbb{R}$.

1. $\int_{\mathbb{R}} \hat{p}_h(x)\,dx = n^{-1}\sum_{i=1}^{n} 1 = 1.$
2. Without the 1/h factor, each bump would instead have area h.
3. Bandwidth changes width and height together, preserving total probability.

Wasserman, 36-708 Density Estimation, §4; original example

EXERCISE

Kernel density estimation

What does a larger bandwidth change?

Hold $x_i \in \{-2.2, -2, -1.8, 1.8, 2, 2.2\}$ fixed. Increase h for a Gaussian kernel K of area 1. What changes?

$$\hat{p}_h(x) = \frac{1}{nh} \sum_i K((x - x_i)/h)$$

`x,x_i` query location and fixed observations

`n,h` sample count and positive bandwidth

A Wider bumps, same total area

B More total probability mass

C Narrower bumps and more sharp peaks

Wasserman, 36-708 Density Estimation, §4; original example

EXAMPLE

Kernel density estimation

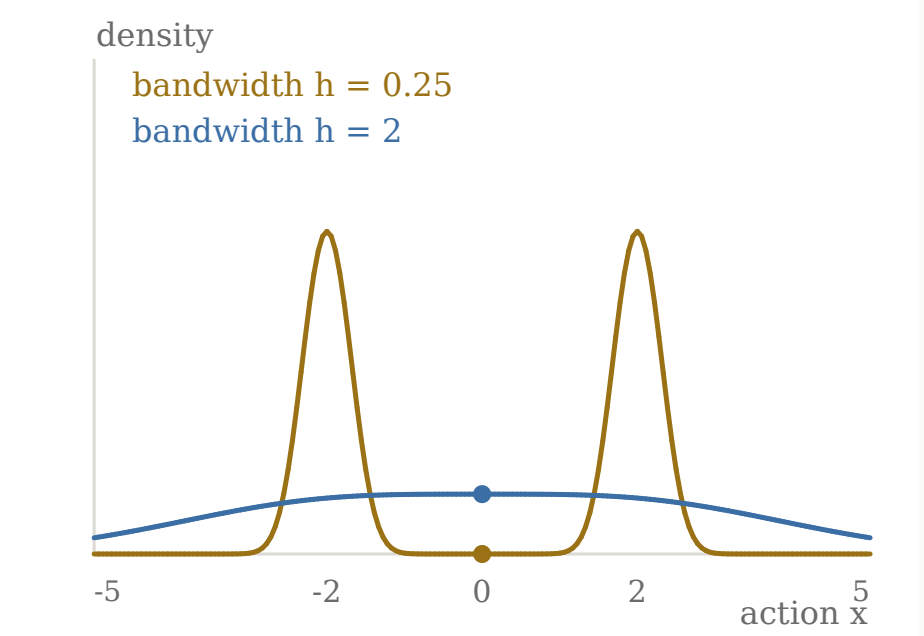
The same actions support different apparent shapes

COURSE-CREATED WORKED EXAMPLE

A density mode maximizes density height. Estimate it from actions ±1.8,±2,±2.2 using Gaussian kernels with bandwidths 0.25 and 2.

$$\hat{m}_h \in \arg\max_x \hat{p}_h(x)$$

- $\hat{p}_h(x)$ estimated density at action x
- $\hat{m}h$ a maximizing action
- h bandwidth in action units



Wasserman, 36-708 Density Estimation, §4; original example

EXERCISE

Density and modes

Choose an action from two observed groups

Actions form two separated groups around -2 and 2. A robot needs one plausible action.

A. Always execute the mean 0

B. Model the distribution and check the sampled action

C. Declare every observed value a true mode

Wasserman, 36-708 Density Estimation, §§1,3–4

DEFINITION

Stable probability computation

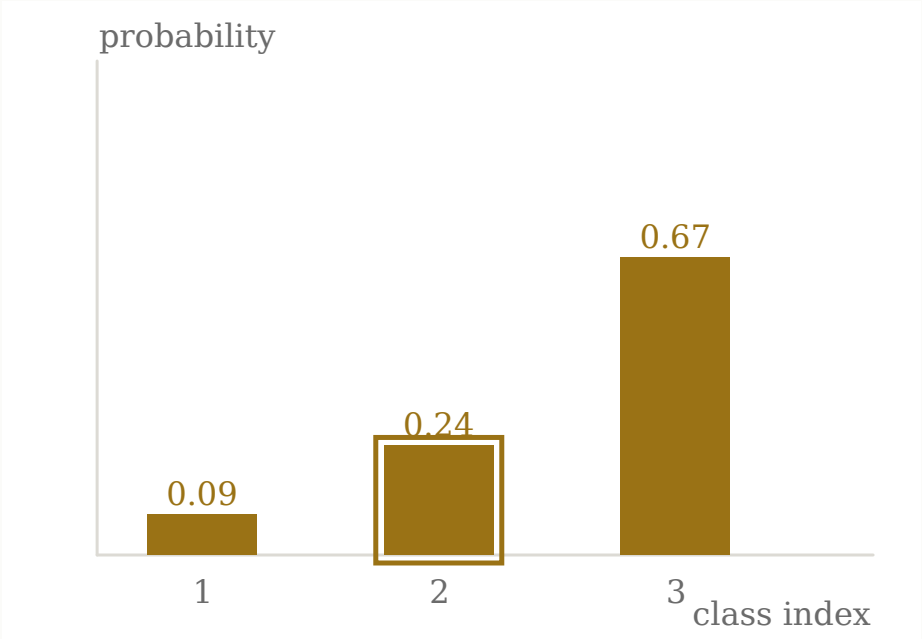
Exponentials turn scores into probabilities

Softmax converts finite real scores, called logits, into positive class probabilities that sum to one.

$$q_i = \frac{e^{z_i}}{\sum_{j=1}^{K} e^{z_j}}$$

`z_i` finite real logit for class i

`q_i,K` class probability and number of classes



Blanchard, Higham & Higham, arXiv:1909.03469, §§1,4

DERIVATION

Stable probability computation

A shared shift leaves softmax unchanged

Multiply numerator and denominator by e^{-c}. Choose $c=\max_j z_j$.

$$q_i = \frac{e^{z_i - c}}{\sum_j e^{z_j - c}}$$

`z_i,z_j` finite scores for classes i,j

`q_i` probability for class i

`c` maximum score

Blanchard, Higham & Higham, arXiv:1909.03469, §§1,4

Stable probability computation

A shared shift leaves softmax unchanged

Cancel a common exponential factor

z_i is the finite score for class i, q_i its probability, and c any finite shared constant.

1. $q_i = \frac{e^{z_i} e^{-c}}{\sum_j e^{z_j} e^{-c}}$.
2. $= \frac{e^{z_i - c}}{\sum_j e^{z_j - c}}$ by exponential multiplication.

Blanchard, Higham & Higham, arXiv:1909.03469, §§1,4

Stable probability computation

A shared shift leaves softmax unchanged

Choose the shift for numerical safety

Set $c=\max_j z_j$.

1. Every shifted score satisfies $z_i-c\le 0$.
2. Therefore $0<\exp(z_i-c)\le 1$ in exact arithmetic; at least one equals 1.
3. This avoids exponential overflow. Tiny terms can still underflow in finite precision.

Blanchard, Higham & Higham, arXiv:1909.03469, §§1,4

EXAMPLE

Stable probability computation

Large logits can encode ordinary probabilities

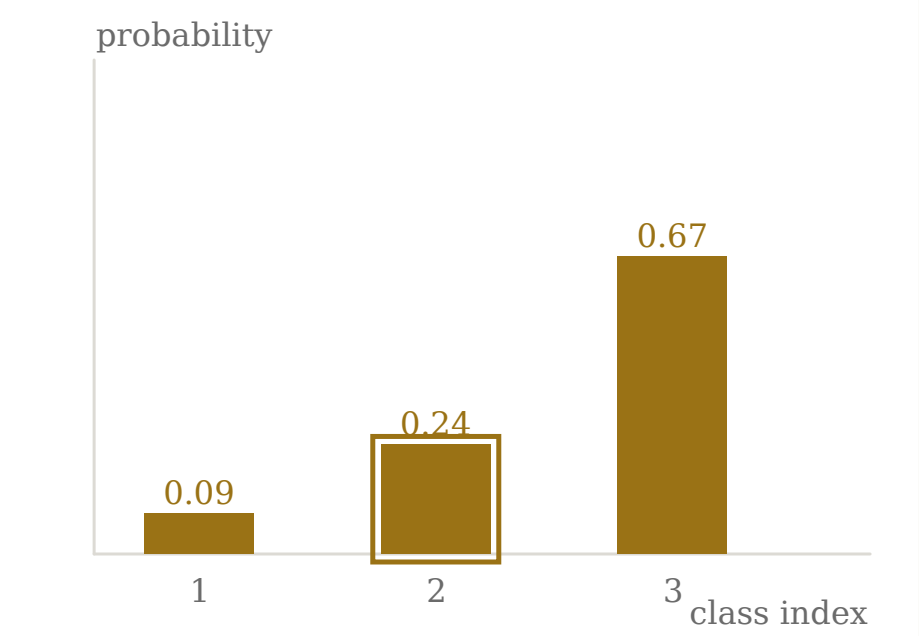
COURSE-CREATED WORKED EXAMPLE

Given z=(1000,1001,1002), subtract c=1002.

$$q = \frac{(e^{-2}, e^{-1}, 1)}{e^{-2} + e^{-1} + 1} \approx (0.090, 0.245, 0.665)$$

`z,q` logit vector and probability vector

`c` maximum logit 1002



Blanchard, Higham & Higham, arXiv:1909.03469, §§1,4

EXERCISE

Stable probability computation

Which change preserves the distribution?

For K real scores z, which change always preserves softmax in exact arithmetic? Here 1 is the K-vector of ones.

A $z \to z + 1000 \cdot 1$

B $z \to 2z$

C $z \to -z$

Blanchard, Higham & Higham, arXiv:1909.03469, §§1,4

DERIVATION

Stable probability computation

Log-sum-exp keeps the normalization in log space

Log-sum-exp (LSE) is the natural logarithm of a sum of exponentials. Evaluate it using the largest finite logit c.

$$\mathrm{LSE}(z) = c + \log \sum_j e^{z_j - c}, \quad c = \max_j z_j$$

- `z` vector of finite class scores
- `c` maximum score
- `LSE(z)` log of the sum of exponentials

Blanchard, Higham & Higham, arXiv:1909.03469, §§1,4

Stable probability computation

Log-sum-exp keeps the normalization in log space

Factor before taking the logarithm

$c = \max_j z_j$ for finite logits z.

1. $\sum_j e^{z_j} = e^c \sum_j e^{z_j - c}$.
2. $\log \sum_j e^{z_j} = \log e^c + \log \sum_j e^{z_j - c}$.
3. $\mathrm{LSE}(z) = c + \log \sum_j e^{z_j - c}$.

Blanchard, Higham & Higham, arXiv:1909.03469, §§1,4

Stable probability computation

Log-sum-exp keeps the normalization in log space

Track the constant

LSE is a log normalization value, not a probability ratio.

1. Adding c to every logit adds c to LSE.
2. The shift term cancels in softmax ratios but must be restored in LSE.
3. For z=(1000,1000), LSE=1000+log 2.

Blanchard, Higham & Higham, arXiv:1909.03469, §§1,4

DERIVATION

Stable probability computation

Compute log probabilities before exponentiating

For a known class y, use the log probability directly in cross entropy.

$$\log q_i = z_i - \mathrm{LSE}(z), \qquad \ell = -\log q_y$$

`z_i,q_i` score and probability for class i

`LSE(z)` log-sum-exp of the scores

`y` observed class

ℓ negative log loss in nats

Blanchard, Higham & Higham, arXiv:1909.03469, §§1,4

Stable probability computation

Compute log probabilities before exponentiating

Take the log of the softmax ratio

For class i, z_i is its score and q_i its probability. $\mathrm{LSE}(z)=\log\Sigma_j\exp(z_j)$.

1. $\log q_i = \log e^{z_i} - \log \sum_j e^{z_j} = z_i - \mathrm{LSE}(z)$.
2. For the observed class y, $\ell = -\log q_y$.

Blanchard, Higham & Higham, arXiv:1909.03469, §§1,4

Stable probability computation

Compute log probabilities before exponentiating

Keep the computation in shifted form

$c = \max_j z_j$ and y is the observed class index.

1. $\log q_i = (z_i - c) - \log \sum_j e^{z_j - c}$.
2. $\ell = \log \sum_j e^{z_j - c} - (z_y - c)$.
3. Average ℓ across observations to obtain empirical cross entropy in nats.

Blanchard, Higham & Higham, arXiv:1909.03469, §§1,4

EXERCISE

Stable probability computation

Compute the observed-class log loss

z=(1000,999). The observed class is the second. Compute loss in nats.

$$\ell = \log \sum_j e^{z_j - c} - (z_y - c), \quad c = \max_j z_j$$

y observed class index; log is natural

A About 0.313

B About 1.313

C Infinite because e^{1000} overflows

Blanchard, Higham & Higham, arXiv:1909.03469, §§1,4

DERIVATION

Stable probability computation

The gradient of log-sum-exp is softmax

For K classes, $z, q \in \mathbb{R}^K$ are score and probability columns. The one-hot target e_y has a 1 at observed class y and 0 elsewhere.

$$\frac{\partial}{\partial z_i} \log \sum_j e^{z_j} = \frac{e^{z_i}}{\sum_j e^{z_j}} = q_i$$

`z`	column vector of logits
`gradient`	column probability vector q

Blanchard, Higham & Higham, arXiv:1909.03469, §§1,4

Stable probability computation

The gradient of log-sum-exp is softmax

Differentiate the scalar normalizer

$S=\Sigma_j \exp(z_j)$, and the logits are independent coordinates.

1. $\partial S/\partial z_i = e^{z_i}$.
2. $\partial \log S/\partial z_i = S^{-1}\partial S/\partial z_i = e^{z_i}/S = q_i$.

Blanchard, Higham & Higham, arXiv:1909.03469, §§1,4

Stable probability computation

The gradient of log-sum-exp is softmax

Differentiate the observed-class loss

ℓ=LSE(z)−z_y; e_y is the one-hot column target.

1. $\nabla_z \operatorname{LSE}(z) = q$.
2. $\nabla_z z_y = e_y$, so $\nabla_z \ell = q - e_y$.
3. The gradient components sum to 1−1=0, consistent with shared-shift invariance.

Blanchard, Higham & Higham, arXiv:1909.03469, §§1,4

EXERCISE

Stable probability computation

The loss gradient changes the right scores

q=(0.7,0.2,0.1). The true class is 2. Use gradient descent on logits.

$$\nabla_z \ell = q - e_2, \qquad z_{new} = z - \eta \nabla_z \ell$$

`z, z_new` logits before and after the update

ℓ loss

`e_2` one-hot target (0,1,0)

η positive step size

A. Increase all logits equally

B. Increase class 2, decrease classes 1 and 3

C. Decrease class 2, increase classes 1 and 3

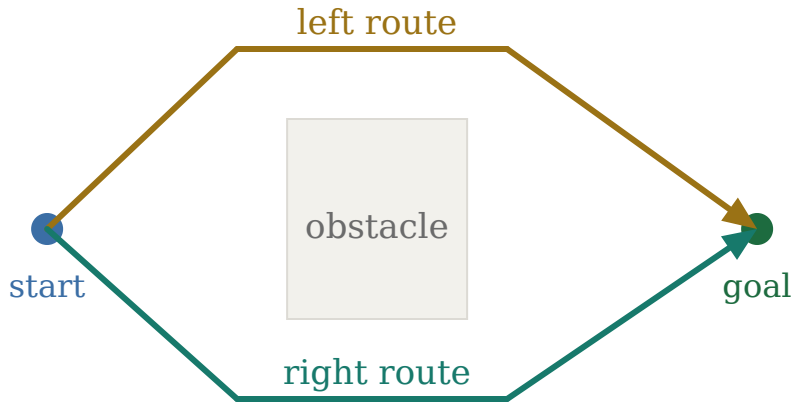
Blanchard, Higham & Higham, arXiv:1909.03469, §§1,4

SETUP

Robot synthesis

Choose the quantities before calculating them

Source p=(0.8,0.2), model q=(0.6,0.4). Plan 100 independent trials. Find entropy H(p), cross entropy H(p,q), KL from p to q, and the left-route fraction’s standard error.



Goodfellow et al., Deep Learning (2016), §3.13

SOLUTION

Robot synthesis

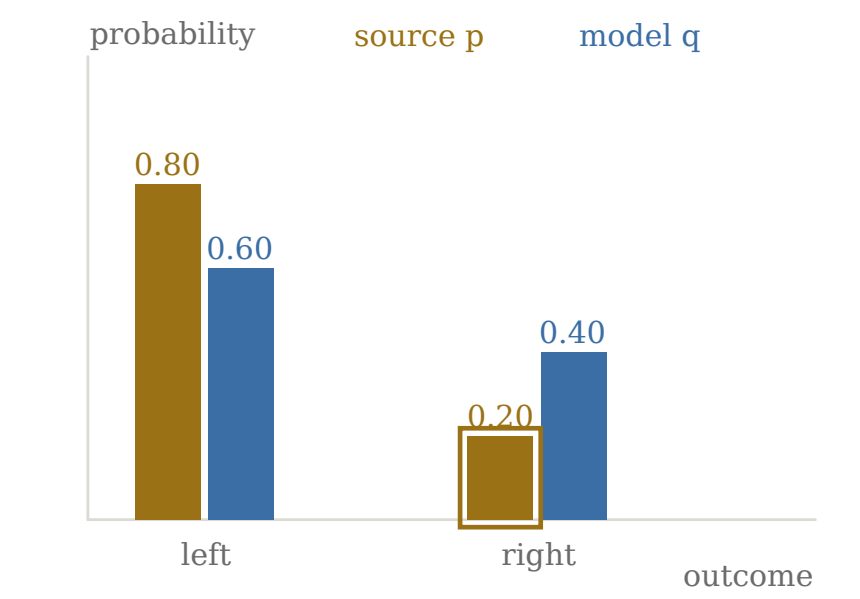
Information and sampling answer different parts

COURSE-CREATED WORKED EXAMPLE

Source p=(0.8,0.2), model q=(0.6,0.4), both left/right. Fix n=100 independent trials.

$$H(p) \approx 0.722, \quad H(p,q) \approx 0.854, \quad D_{\mathrm{KL}}(p\|q) \approx 0.132$$

- `H(p)` source entropy
- `H(p,q)` cross entropy
- `D_KL` p-to-q divergence (all in bits)
- `SE` standard error of left-route fraction: √(0.8·0.2/100)=0.04



Goodfellow et al., Deep Learning (2016), §3.13

EXERCISE

Robot synthesis

Evaluate a changed route frequency

In left/right order, source p changes from (0.8,0.2) to (0.2,0.8). Model q stays (0.6,0.4). Compare entropy and cross entropy.

A. Entropy stays the same; cross entropy rises

B. Both quantities stay the same

C. Entropy must increase

Goodfellow et al., Deep Learning (2016), §3.13

RECAP

The reasoning path

From scoring predictions to generating actions

Probabilities support log losses. Samples support estimates. Structured density models support multiple action choices.

Score predictions	Cross entropy and KL
Measure sensor information	Mutual information
Use trial evidence	Sample mean and standard error
Model action groups	Histogram and kernel density
Compute log scores	Softmax and log-sum-exp

Goodfellow et al., Deep Learning (2016), §3.13

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Probability and information

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